

Smart bus tracking and monitoring system using camera

Final Presentation

Group No: 6

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Abstract

This project presents a smart bus attendance system using face recognition and GPS tracking to automatically identify students and record attendance in real time. It is cost-effective and improves efficiency, reliability, and safety.

Introduction

- This project presents a mobile-based bus attendance system that uses face recognition and GPS to track students boarding and exiting the college bus.
- The solution is cost-effective, easy to deploy, and requires only a single smartphone without any additional hardware.

Problem Statement

- Ensuring student safety and accurate bus attendance is challenging with manual methods and RFID systems due to errors, delays, and hardware dependency, creating a need for a low-cost, automated real-time solution.

Literature Review

Base Paper	Author	Methodology	Limitation
Smart College Bus Tracking: Real-Time Location & ETA Prediction, IJNRD, 2025	M. Soundharya et al.	Live location tracking, ETA prediction	GPS accuracy and internet dependency
Real-Time College Bus Tracking and Student Attendance System, IJNRD, 2025	R. Anjana et al.	GPS tracking, automated attendance, alerts	Lighting issues, privacy concerns
Attendance System Using Facial Recognition (IJERT), May 2023	Tushar Rane, Sagar Dhoot, Nayan Jain, Ramesh Naphade, Rohit Virani	Image capture, attendance marking, email notification	No automated notification system
Bus Attendance System through Facial Recognition by Implementing AI Integration, IJNRD, Nov 2022	Johnson C., Karthikeyan S., Sakthivel R., Shankar K., Sakthi A.	Facial recognition, base64 encoding, student database, attendance marking, notification	Energy-efficient hardware, biometric integration issues, AI integration with student performance data
Bus Monitoring System (IJERT), April 2021	Neha Parkara, Ashwini Patil et al.	Bus tracking using smart connectivity	Additional hardware required, privacy concerns

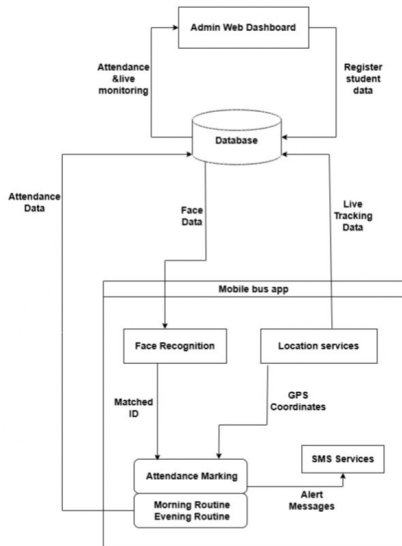
Proposed System

- Automatic attendance using facial recognition.
- Live tracking of college buses.
- Centralized storage of attendance and location data.
- Easy monitoring for administrators.
- Reduces manual work and errors.

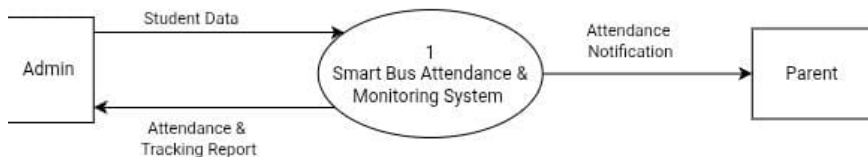
Technologies used

Frontend: Flutter. Backend: Python Flask. Firestore stores student data, embeddings, GPS logs, and attendance. Face detection uses Google ML Kit; recognition uses TensorFlow Lite. Additional integrations: camera processing, Google Maps SDK, Directions API, Geolocation API, and OpenStreetMap for route visualization.

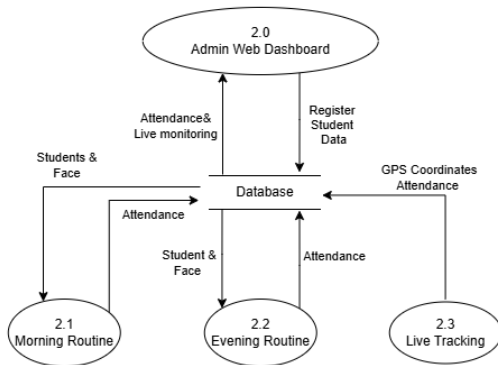
SYSTEM ARCHITECTURE DIAGRAM



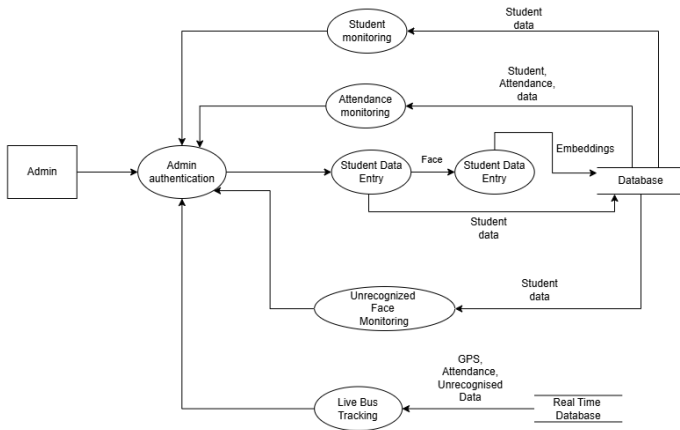
DATAFLOW DIAGRAM LEVEL 0



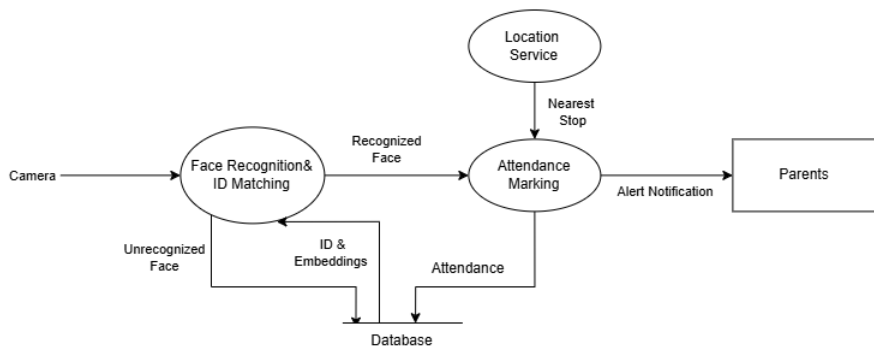
DATAFLOW DIAGRAM LEVEL 1



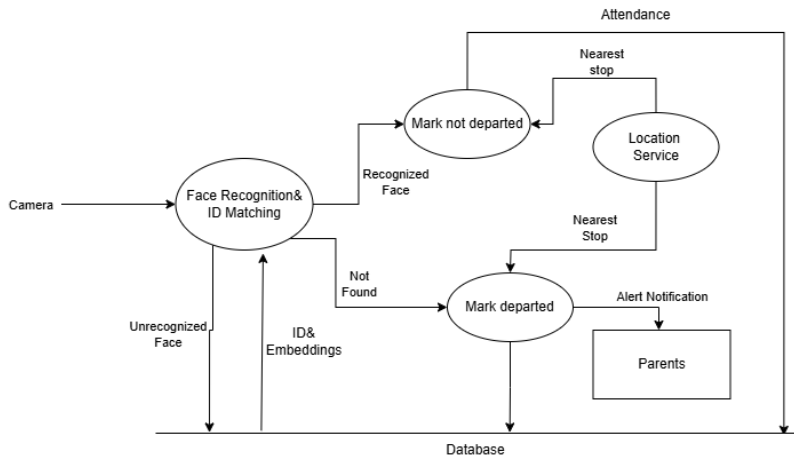
DATAFLOW DIAGRAM LEVEL 2.0



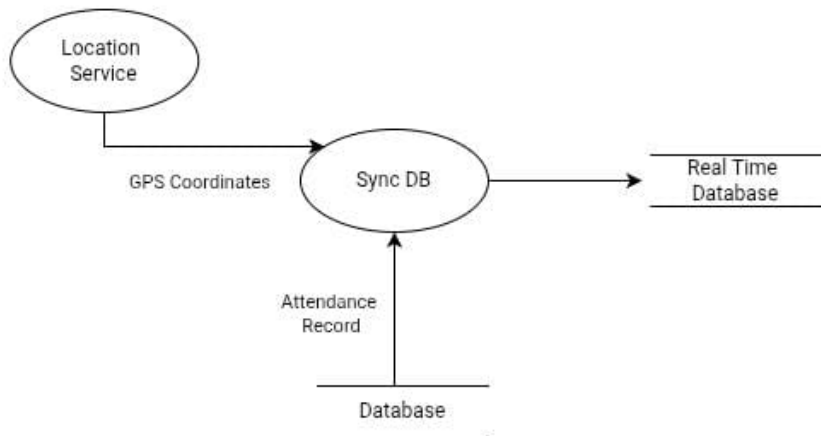
DATAFLOW DIAGRAM LEVEL 2.1



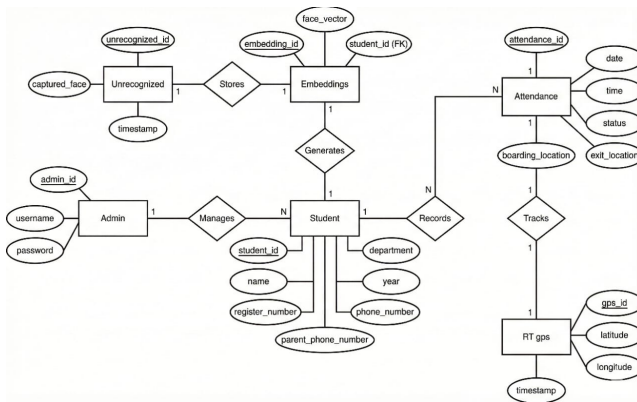
DATAFLOW DIAGRAM LEVEL 2.2



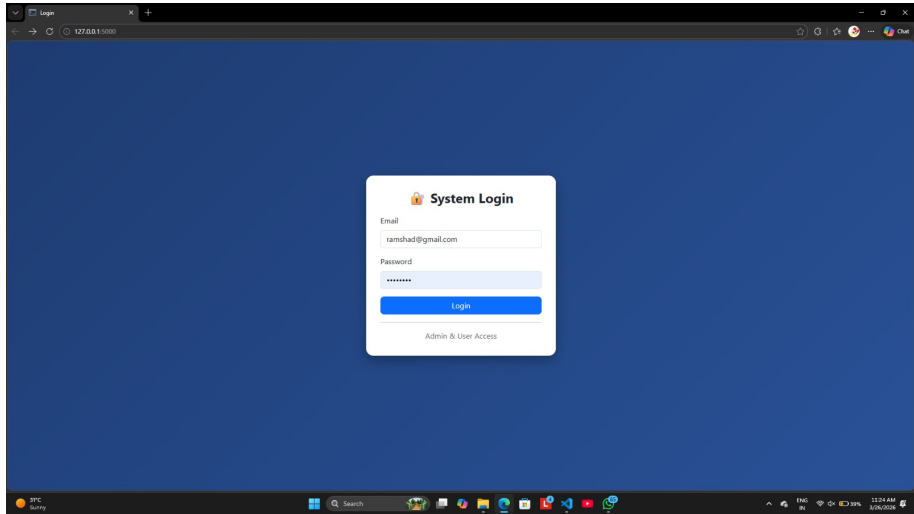
DATAFLOW DIAGRAM LEVEL 2.3



ER Diagram



Demo



Smart Bus Admin

DashboardBusesStudentsAttendanceConfigLogout

Dashboard

Total Buses
1

Total Students
2

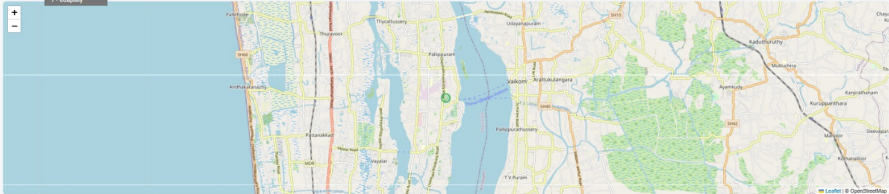
Today's Attendance
0

Live Bus Location

Select Bus1 - odapally

Live tracking-- Select a bus --

1 - odapally



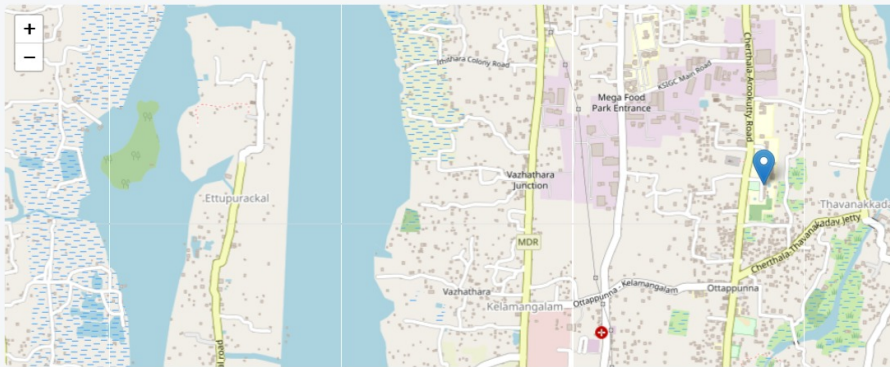
Live Students (On Bus)

No students marked yet

Settings

Set College Location

College Name



Selected: 9.743298, 76.364052

Smart Bus Admin

Add Student

Full Name

akash

Admission Number

2

1 - edapally

Transport Routes

Total Fee: ₹4000

☒ Morning Route

Vyttila (₹2000)

☒ Evening Route

Vyttila (₹2000)

Student Phone Number

1234567890

Parent Phone Number

0987654321

Student Login

Student Email

akash@gmail.com

Student Password

Parent Login

Parent Email

akashparent@gmail.com

Parent Password

Student Photo

 Upload

 Take Photo

Choose File

WhatsApp I... 7.47 PM.jpeg

Add Student

Manage Students

Filter by bus: All buses ▾

Photo	Name	Admission No	Bus	Parent Phone	Status	Actions
	akash	2	1	0987654321	active	Manage <a>Delete
	ramshad	1	1	8113997341	active	Manage <a>Delete
	ravi	3	2	7453672864	active	Manage <a>Delete

Smart Bus Admin

Bus Module



Add Bus

Create a new bus and assign routes



Manage Buses

View, edit, delete existing buses



Manage Drivers

Add, list, and delete drivers

Smart Bus Admin

Add Bus

Bus Name

Bus Number

Student Limit

Bus Permit Image

 WhatsApp Image 2026-02-26 at 8.32.07 PM.jpeg

Add Bus

Smart Bus Admin						Dashboard	Buses	Students	Attendance	Config	Logout
Manage Buses											
Name	Number	Limit	Strength	Status	Actions						
cherthala	2	40	0	inactive	Manage	<button>Delete</button>					
edapally	1	40	2	inactive	Manage	<button>Delete</button>					

Manage Bus

Name

cherthala

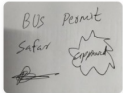
Bus Number

2

Student Limit

40

Bus Permit Image



Change Permit

Leave empty to keep existing permit.

Assigned Driver

Not Assigned

Status

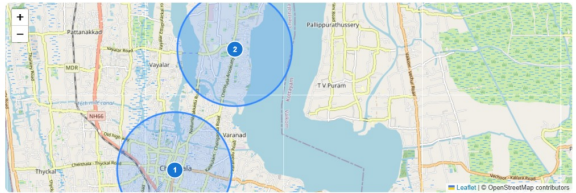
Inactive

Update Bus

Manage Route

Morning

Evening



Morning Stops

Stop 1

Stop Name Cherthala KSRTC Road

Radius: 2000m

Fee (₹) 2000

Remove

Stop 2

Stop Name Kochuramapuram

Radius: 2000m

Fee (₹) 1000

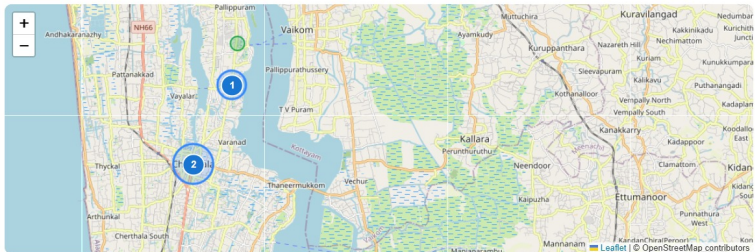
Remove

Save Morning Route

Manage Route

 Morning

 Evening



Evening Stops

Stop 1

Stop Name Radius: 770m Fee (₹) Remove

Stop 2

Stop Name Radius: 1040m Fee (₹) Remove

 Save Evening Route

View Students

Assign Driver

Students Assigned

Student list will load here...

akash

2

ramshad

1

View Students

Assign Driver

 **Select Existing Driver**

Select Driver

Select Driver

Assign Driver

 **Add New Driver**

rajesh

2453675463

rajesh@gmail.com

Add Driver



Select Existing Driver

rajesh (2453675463)



Assign Driver

Manage Bus

Name

cherthala

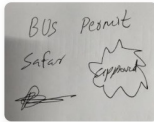
Bus Number

2

Student Limit

40

Bus Permit Image



Change Permit

Leave empty to keep existing permit.

Assigned Driver

rajesh

Status

Inactive

Update Bus

Manage Drivers

+ Add New Driver

Driver Name

Driver Phone

Driver Email

Driver Password

Add Driver

Name	Phone	Email	Status	Assigned Bus	Actions
sumesh	2341679221	sumesh@gmail.com	assigned	1	Unassign Delete
rajesh	2453675463	rajesh@gmail.com	assigned	2	Unassign Delete



Parent Portal

Real-Time Bus Tracking System

EMAIL ADDRESS

 guru@gmail.com

PASSWORD



 Signing In...

 Secured by Firebase Authentication

Hello, ravi ✨

Your child's real-time bus tracking dashboard

Active

Student Information
Personal details & contact



ravi

3

Roll: 3

5432163452

7453672864

Bus & Driver
Assigned vehicle details

BUS NUMBER

2

STATUS

Inactive

OCCUPANCY

1/40 students

DRIVER

rajesh

2453675463

BUS PERMIT

Route Information
Stop schedule and fees

Morning Evening

#	STOP NAME	FEE	COORDINATES
1	Cherthala KSRTC Road	₹2000	9.6857, 76.3422
2	Kochuramapuram	₹1000	9.7234, 76.3613



Fee Records

Payment status and history



No fee records available.

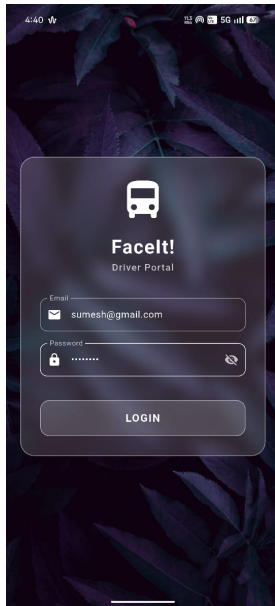


Today's Attendance

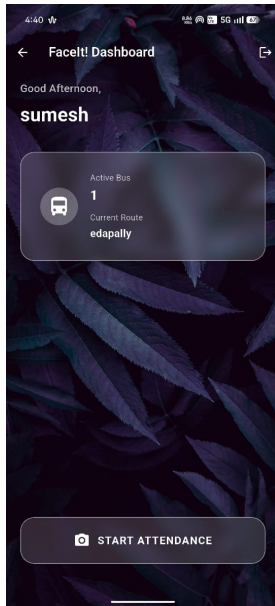
Live scan record



No attendance record for today.

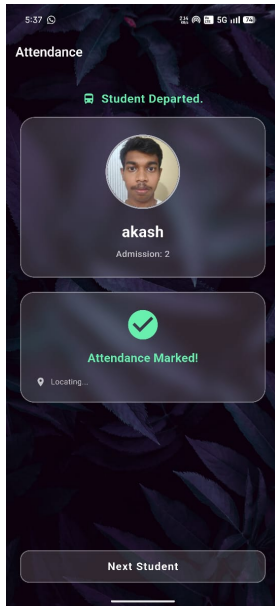


Demo



Demo





- The system can be improved by adding a mobile app for parents and staff to check attendance and track the bus in real time.
- The accuracy of face recognition can be enhanced to work better in different lighting conditions and with masks.
- The system can also include alert notifications to inform parents when students board or leave the bus.
- Additionally, it can be connected with the college management system for easier data handling and reporting.

Conclusion

Implemented to automate student attendance using face recognition. It eliminates manual errors and ensures real-time, accurate identification by integrating face detection, embedding, and recognition techniques. Attendance is stored with date and time in a database, making the system reliable and easy to manage. Overall, the system is efficient, time-saving, and demonstrates the practical use of computer vision in real-world applications.

References

- ❶ Parkara, N., Patil, A., et al., "Bus Monitoring System," International Journal of Scientific and Academic Research in Technology (IJSART), 2021.
- ❷ Rane, T., Dhoot, S., Jain, N., Naphade, R., Virani, R., "Attendance System Using Facial Recognition," International Journal of Creative Research Thoughts (IJCRT), 2023.
- ❸ Johnson, C., Karthikeyan, S., Sakthivel, R., Shankar, K., Sakthi, A., "Bus Attendance System Through Facial Recognition by Implementing AI Integration," International Journal of Novel Research and Development (IJNRD), 2023.
- ❹ Aruna, R., Laksha, M., Prakashini, P., Arun, S., "Real-Time College Bus Tracking and Student Attendance System," International Journal of Scientific Research and Engineering Development (IJSRED), 2025.
- ❺ Sowndharya, M., Kavya, R., Karthik, S., Ananya, S., "Smart College Bus Tracker: Real-Time Location and ETA Prediction," International Journal of Novel Research and Development (IJNRD), 2025.